PROBLEM 24

Particle on a Circumference with a Spring

Original statement · Problems.pdf, p. 12

Wire & Elastic Potential

A heavy particle P is constrained to move along a circumference and it is attached to a spring as shown in the figure. The mass of the particle is m, and the radius of the circumference is a. The other end of the spring is attached to a point of the y axis with coordinates $(0, \beta a)$ and the constant of the spring is equal to k.

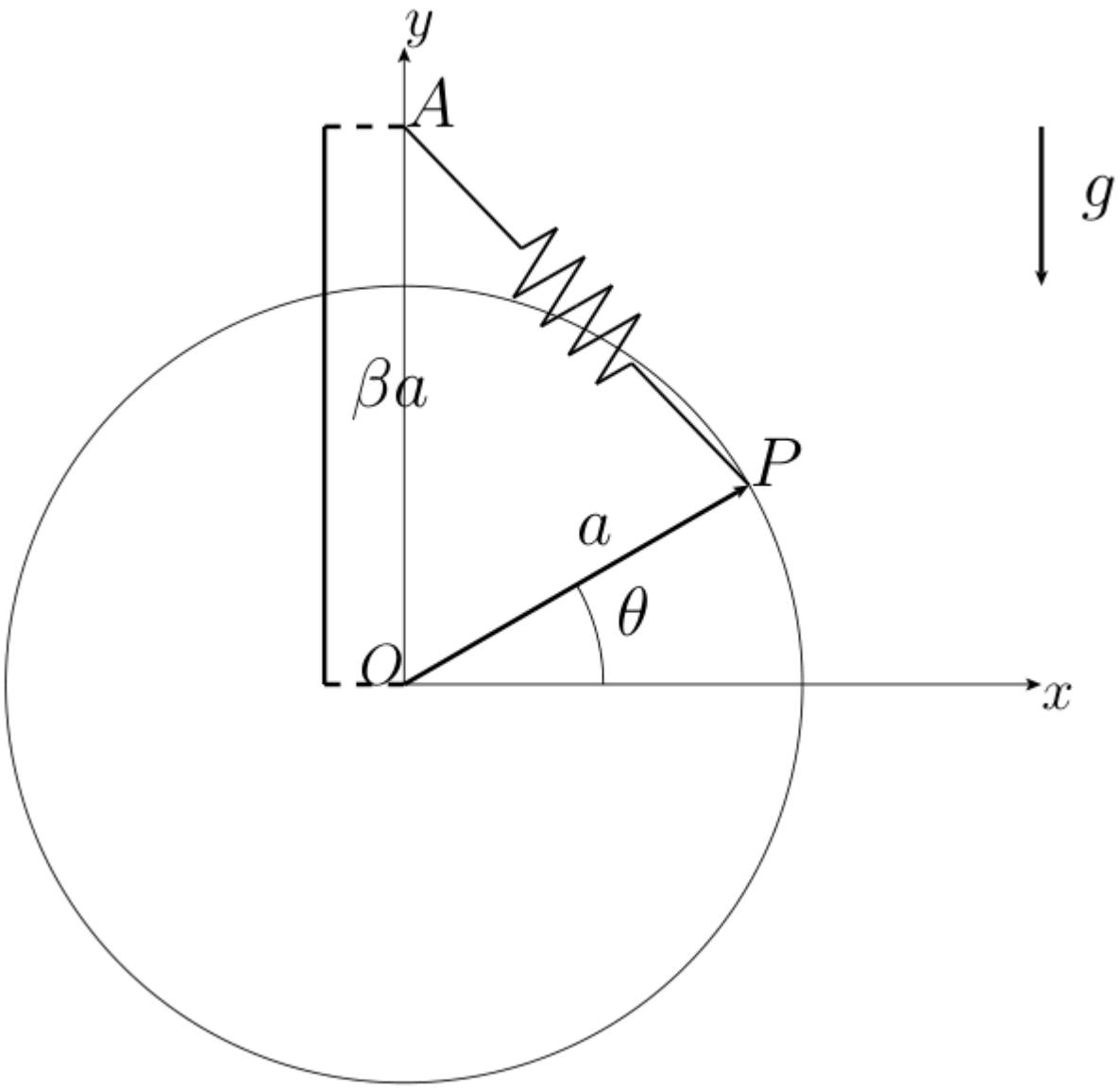


Figure: Particle P on the circumference of radius a centred at O, spring from A = (0, βa) to P, angle θ from the x axis (Problems.pdf, p. 12).

(a) Compute the potential of the forces acting on the particle as a function of the angle θ (shown in the figure).

(b) Identify the equilibrium positions.

The particle departs from the point $(a, 0)$ of the circumference with an initial velocity $\mathbf{v}_0 = v_0\mathbf{j}$.

(c) Reduce the motion to integrals (i.e., quadratures).

(d) If $\beta = \frac{mg}{ka}$, obtain the position of the particle as a function of time.

Viscous Damping Case:
Consider now the situation in which an additional force acts on the particle. The force is proportional to, and in the same direction as, the instantaneous velocity. The constant of proportionality is $2m\mu$. The initial conditions are the same as before.

(e) Obtain the position of the particle as a function of time.

(f) Obtain the reaction of the circumference.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m slides along a vertical circular wire of radius a centered at O. A linear spring of stiffness k and zero natural length connects the particle P to anchor point $A(0, \beta a)$ on the vertical axis.
- **CDOFs:** 1 degree of freedom, angle θ measured counterclockwise from the positive x-axis.
- **Forces:**

1. Gravity: $-mg\mathbf{j}$.

2. Spring force: $\mathbf{F}_s = -k\mathbf{AP}$.

3. Constraint reaction $\mathbf{N}$ normal to the wire.

- **Initial State ($t = 0$):** Released from $(a, 0) \implies \theta(0) = 0$ with $\mathbf{v}_0 = v_0\mathbf{j} \implies \dot{\theta}(0) = v_0/a$.
- **Parameters:** $m, a, k, g, \beta, v_0 > 0$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Position of anchor A: $\mathbf{r}_A = \beta a \mathbf{j}$.
- Position of particle P: $\mathbf{r}_P = a\cos\theta\,\mathbf{i} + a\sin\theta\,\mathbf{j}$.
- Spring elongation vector:

$$\mathbf{AP} = \mathbf{r}_P - \mathbf{r}_A = a\cos\theta\,\mathbf{i} + a(\sin\theta - \beta)\,\mathbf{j}$$

- Length squared of spring:

$$|\mathbf{AP}|^2 = a^2\cos^2\theta + a^2(\sin\theta-\beta)^2 = a^2\left[\cos^2\theta + \sin^2\theta - 2\beta\sin\theta + \beta^2\right] = a^2(1+\beta^2-2\beta\sin\theta)$$

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Total Potential Energy Function $V(\theta)$

> **Pedagogical Justification:** Both gravity and the linear spring are conservative forces. The total potential energy is the sum of gravitational and elastic potentials.

$$V_g(\theta) = mgy = mga\sin\theta$$

$$V_s(\theta) = \frac{1}{2}k|\mathbf{AP}|^2 = \frac{1}{2}ka^2(1+\beta^2-2\beta\sin\theta)$$

Summing both contributions:

$$V(\theta) = mga\sin\theta + \frac{1}{2}ka^2(1+\beta^2) - k\beta a^2\sin\theta = \frac{1}{2}ka^2(1+\beta^2) + (mg - k\beta a)a\sin\theta$$

Part (b): Identification of Equilibrium Positions

Equilibrium requires $\frac{dV}{d\theta} = 0$:

$$\frac{dV}{d\theta} = (mg - k\beta a)a\cos\theta = 0$$

Two cases emerge:

1. If $mg - k\beta a \neq 0$: $\cos\theta = 0 \implies \theta_1 = \frac{\pi}{2}$ (top) and $\theta_2 = -\frac{\pi}{2}$ (bottom).

2. If $\beta = \frac{mg}{ka}$: $\frac{dV}{d\theta} \equiv 0$ for **all** θ. Every point along the circumference is an equilibrium position (neutral equilibrium)!

Part (c): Reduction of Motion to Quadratures

Conservation of mechanical energy:

$$E = \frac{1}{2}ma^2\dot{\theta}^2 + V(\theta) = \frac{1}{2}mv_0^2 + V(0) = \frac{1}{2}mv_0^2 + \frac{1}{2}ka^2(1+\beta^2)$$

$$\dot{\theta} = \pm\sqrt{\frac{v_0^2}{a^2} - \frac{2(mg - k\beta a)}{ma}\sin\theta}$$

$$\int_0^{\theta(t)} \frac{d\theta'}{\sqrt{\frac{v_0^2}{a^2} - \frac{2(mg-k\beta a)}{ma}\sin\theta'}} = t$$

Part (d): Case $\beta = \frac{mg}{ka}$

When $\beta = \frac{mg}{ka}$, the coefficient $(mg - k\beta a) = 0$.

The potential energy is strictly constant: $V(\theta) = \text{constant}$.

The equation of motion reduces to:

$$\ddot{\theta} = 0 \implies \dot{\theta}(t) = \dot{\theta}(0) = \frac{v_0}{a}$$

Integrating with $\theta(0) = 0$:

$$\theta(t) = \frac{v_0}{a}t$$

The particle circles the wire with strictly uniform angular speed!

Part (e): Viscous Damping Case (Damping force $\mathbf{F}_d = -2m\mu\mathbf{v}$)

With $\beta = \frac{mg}{ka}$, the tangential equation of motion is:

$$ma\ddot{\theta} = F_{d,\theta} = -2m\mu(a\dot{\theta})$$

Divide by ma:

$$\ddot{\theta} + 2\mu\dot{\theta} = 0$$

Let $\omega(t) = \dot{\theta}(t)$:

$$\dot{\omega} + 2\mu\omega = 0 \implies \omega(t) = \omega(0)e^{-2\mu t} = \frac{v_0}{a}e^{-2\mu t}$$

Integrate for position $\theta(t) = \int_0^t \omega(t')dt'$ with $\theta(0) = 0$:

$$\theta(t) = \frac{v_0}{a}\int_0^t e^{-2\mu t'}\,dt' = \frac{v_0}{a}\left[-\frac{1}{2\mu}e^{-2\mu t'}\right]_0^t = \frac{v_0}{2\mu a}\left(1 - e^{-2\mu t}\right)$$

Part (f): Reaction of the Circumference

Radial Newton's second law, with N the component of the reaction along $+\mathbf{e}_r$ (outward), $\mathbf{F}_g \cdot \mathbf{e}_r = -mg\sin\theta$ and $\mathbf{F}_s \cdot \mathbf{e}_r = -ka(1 - \beta\sin\theta)$:

$$-ma\dot\theta^2 = N + \mathbf{F}_g\cdot\mathbf{e}_r + \mathbf{F}_s\cdot\mathbf{e}_r$$

$$N(t) = -ma[\dot\theta(t)]^2 + mg\sin\theta(t) + ka(1 - \beta\sin\theta(t))$$

Substitute $\beta = \frac{mg}{ka}$:

$$N(t) = ka - ma\left(\frac{v_0}{a}\right)^2 e^{-4\mu t} = ka - \frac{mv_0^2}{a}e^{-4\mu t}$$

Erratum: The first version of part (f) had the sign of the reaction wrong ($N = ma\dot\theta^2 + mg\sin\theta + ka(1-\beta\sin\theta)$, which follows from no sign convention). With N the outward component, $N = -ma\dot\theta^2 + mg\sin\theta + ka(1-\beta\sin\theta)$, i.e. $N\,\mathbf{e}_r = ka - (mv_0^2/a)e^{-4\mu t}$ for $\beta = mg/(ka)$ (inward component $-N$). Numeric check: sympy solving $m\mathbf{a} = \mathbf{F}_g + \mathbf{F}_s + N\mathbf{e}_r$ reproduces the general expression with zero residual.

PHASE 4

Physical Interpretation & Dimensional Verification

- As $t \to \infty$, the particle comes to rest asymptotically at $\theta_\infty = \frac{v_0}{2\mu a}$, and the reaction force approaches the static spring preload ka.

Source: Problems.pdf (Page 12)